FRONTEND DEVELOPMENT FOR INTERACTIVE APPLICATIONS | 03/OCTOBER/2025

# PURPOSEFUL MOTION DESIGN

## – Component Library

Click here to view webpage with all components

**PRESENTED BY:**
Mohammad Hilal Malik

**STUDENT ID:**
100472791

# Agenda

| | |
|---|---|
| 03 | Introduction / Rationale |
| 04 | Motion Design Principles in Use |
| 05 | Components Overview |
| 06 | Storyboard Applications |
| 07 | Competitor Research |
| 08 | Conclusion |

# Overview



**The Purposeful Motion Design Component Library** was built to show how **motion enhances user experience by adding clarity, feedback, and delight to everyday interactions**. Motion is used here as a functional tool, not a decoration guiding users and reducing uncertainty.

All components are implemented in **HTML, CSS, and JavaScript**, making them reusable and easy to integrate into real-world interfaces.

My design philosophy:

- **Clarity** – animations support understanding.
- **Feedback** – loaders, checks, and shakes confirm system status.
- **Delight** – small celebratory moments keep users engaged.
- **Predictability** – consistent timing and easing build trust.

**This library demonstrates how thoughtful motion design can make interfaces both usable and enjoyable.**

# Motion Design Principles in Use



This library applies core motion design principles to create animations that feel natural, purposeful, and supportive of user experience.

**Timing**

- **Short transitions** (0.3–0.5s) for inputs and cards,
- **longer cycles** (1s+) for loaders.

**Easing**

- **Ease-in-out** for natural movement.
- L**inear** for continuous loaders.
- **Ease-out** for celebratory effects.

**Feedback**

- Checkmarks, confetti, loaders, and shakes provide **instant responses** to user actions.

**Staging**

- Animations play in sequence (e.g., circle → checkmark → confetti) to **guide attention step by step.**

Together, these principles ensure the animations are consistent, intuitive, and aligned with professional UI/UX standards.

# Components Overview



The library includes eight reusable motion components, each designed with specific timing, easing, and user scenarios in mind.

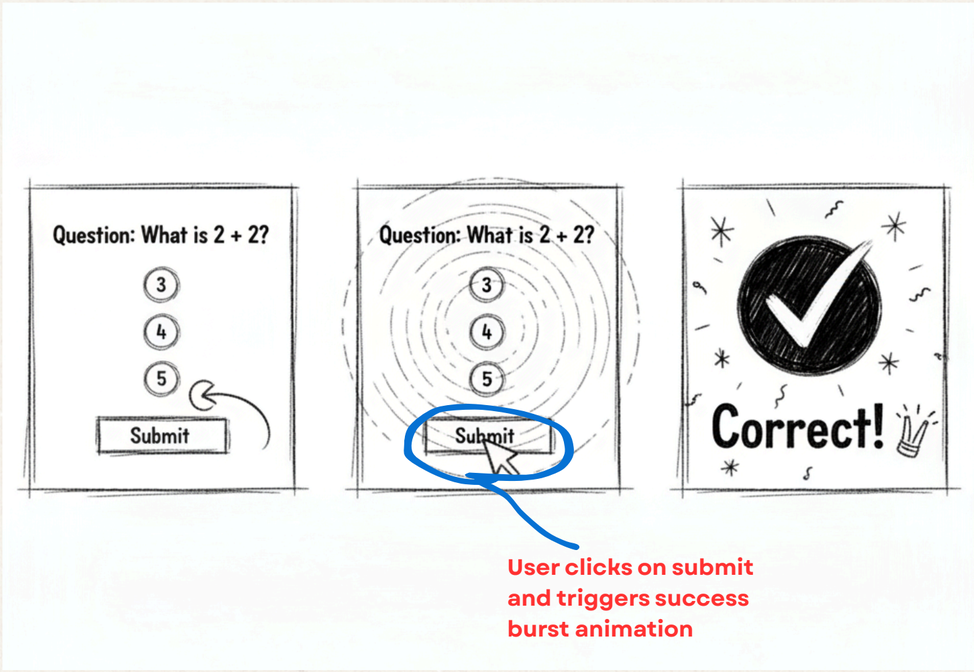
- **Loader** – 1s linear loop · shows background tasks.
- **Success** (Circle + Check + Confetti) – 0.3s + 0.3s + 1s ease-out · celebrates completion.
- **Error / Warning** – 0.5s ease-in-out shake · highlights invalid input.
- **Toggle Switch** – 0.3s ease slide · smooth ON/OFF states.
- **Floating Input** – 0.3s ease · keeps labels visible while typing.
- **Expandable Card** – 0.4s ease expand · reveals hidden content.
- **Modal Transition** – 0.5s ease fade/scale · focuses attention.
- **Drag & Drop** – smooth transforms · rearranges items intuitively.

# Storyboard Applications



### Case 1: Quiz Completion Success

- **Context:** User finishes a quiz and clicks **“Submit.”**
- **Animation Used:** Success circle with checkmark + falling confetti.
- **Timing & Easing:** Circle and checkmark appear in 0.3s ease; confetti falls in 1s ease-out.
- **UX Benefit:** Adds delight and reinforces positive feedback for task completion.
- **Visual:** Success storyboard sketch.

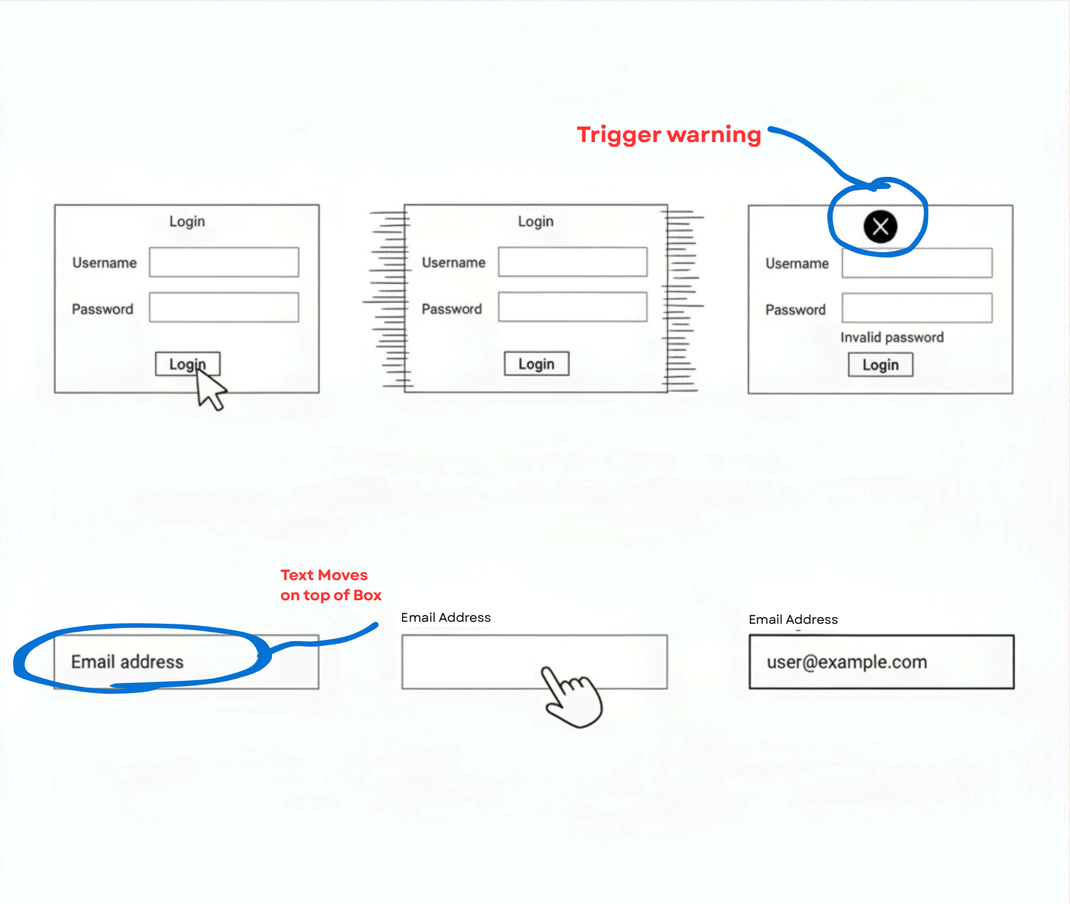


# Storyboard Applications



### Case 2: Form Validation Error & Floating Input

- **Context:** User clicks “Login” with an empty password field.
- **Animation Used:** Error circle shakes (0.5s ease-in-out). Floating input label animates upward when field is active.
- **Timing & Easing:** Error shake 0.5s ease-in-out; floating label 0.3s ease.
- **UX Benefit:** Highlights invalid input instantly; floating label improves clarity while typing.
- **Visual:** Error + floating input storyboard sketch.

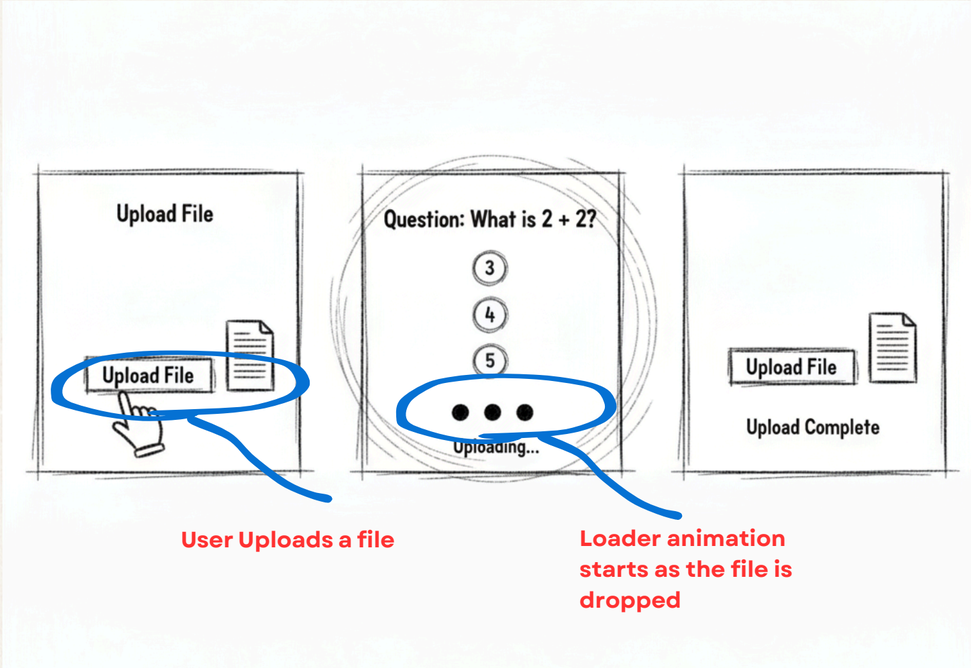


# Storyboard Applications



### Case 3: Loading Task Feedback

- **Context:** User uploads a large file and waits for processing.
- **Animation Used:** Loader with three pulsing dots.
- **Timing & Easing:** Loader cycles every 1s linear.
- **UX Benefit:** Reassures users by showing progress, reducing uncertainty during wait time.
- **Visual:** Loader storyboard sketch.



# Competitor Research

**Duolingo**

Language Learning App

**Headspace**

Mindfulness app

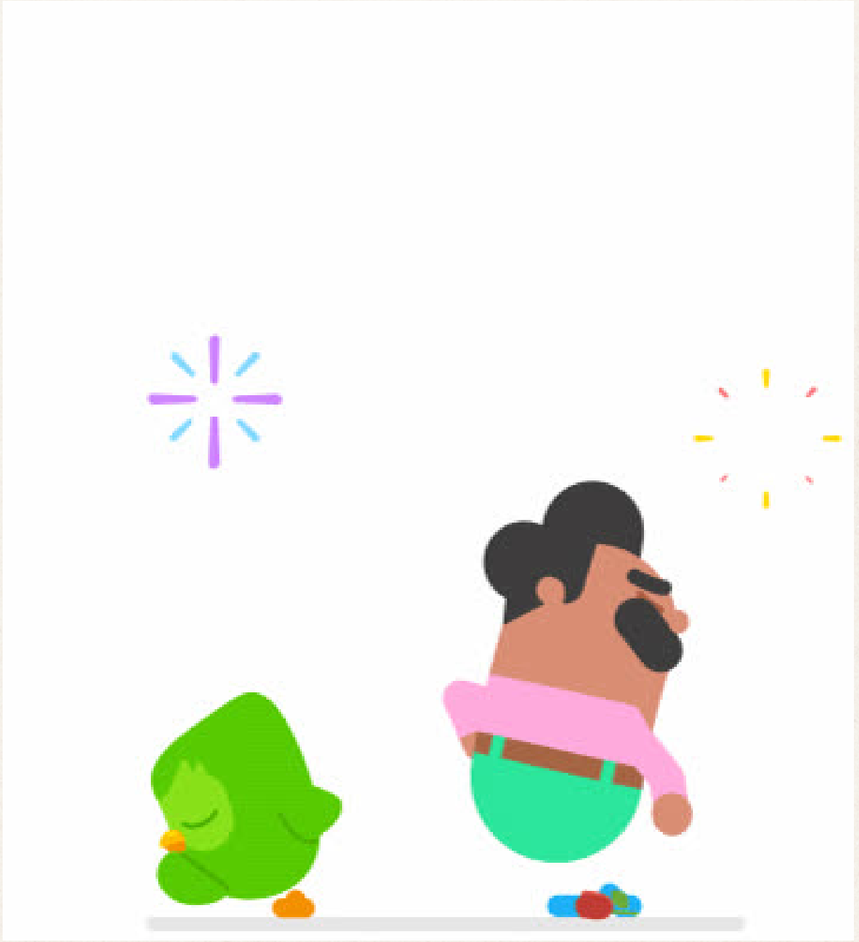
**Todoist**

Productivity app

# Duolingo



A gamified language learning app that helps users practice reading, writing, listening, and speaking in over 40 languages. **Uses streaks, levels, and rewards to motivate daily learning.**



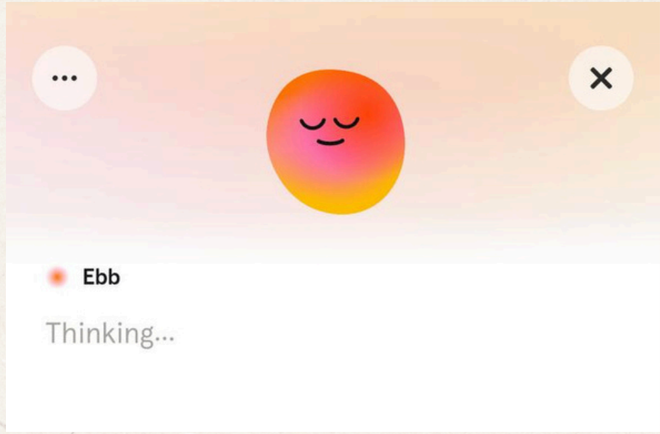
- **Motion Use:** Streak flames, celebratory confetti, character reactions.
- **Timing & Easing:** Fast, bouncy, playful motion.
- **UX Benefit:** Keeps users motivated with game-like micro-celebrations.
- **Takeaway:** Motion design can turn repetitive learning into a rewarding experience.

# Headspace

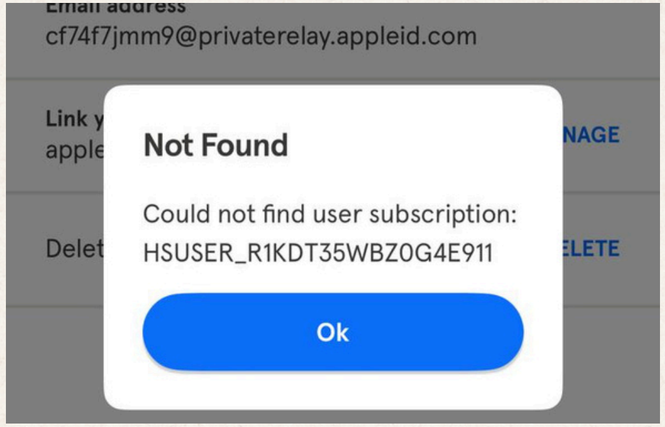


A meditation and mindfulness app that guides users through relaxation, stress reduction, and sleep improvement.
**Offers animations, stories, and breathing exercises to promote calmness.**

- **Motion Use:** Calming character animations, smooth breathing guides.
- **Timing & Easing:** Slow ease-in-out transitions, long cycles (2–3s).
- **UX Benefit:** Reinforces a sense of calm and mindfulness.
- **Takeaway:** Motion can support brand values and user emotional states.




**Thinking Loader**



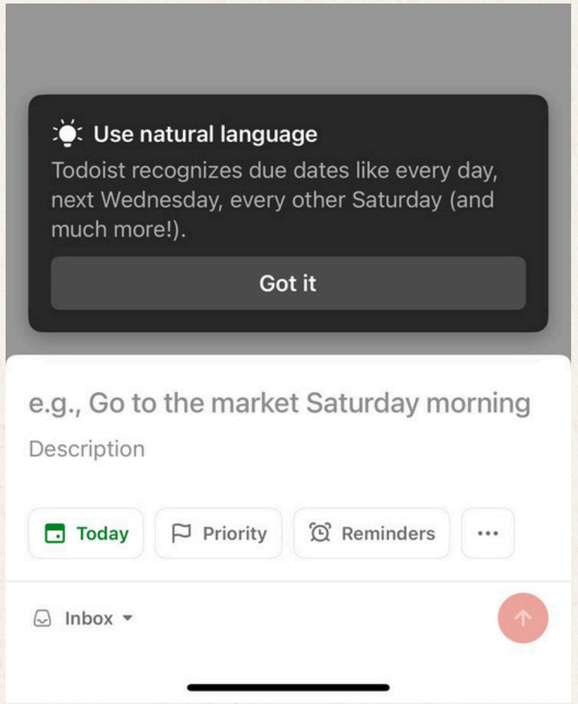

**Modal**

# Todoist

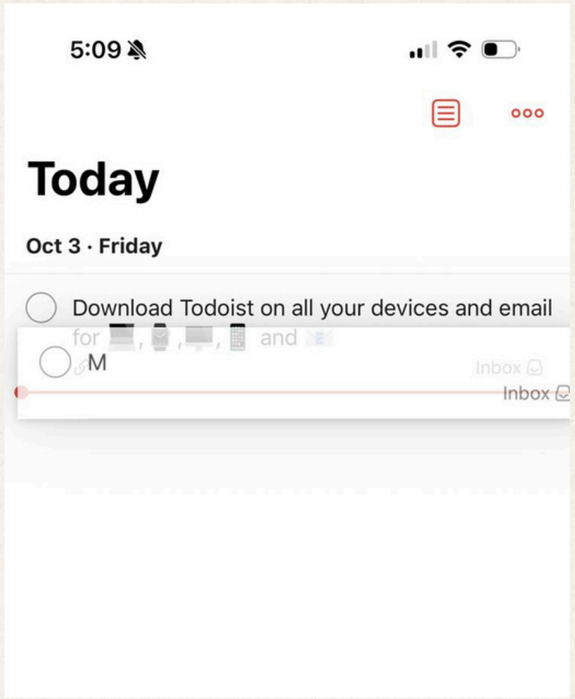


A task management and productivity app for organizing to-do lists, projects, and reminders.
**Popular with professionals and students for keeping track of tasks.**

- **Motion Use:** Subtle checkmark animation when completing tasks.
- **Timing & Easing:** 0.3s ease, quick but smooth.
- **UX Benefit:** Confirms task completion instantly without breaking flow.
- **Takeaway:** Small, subtle animations can create satisfaction without distraction.




Modal




Drag and Move

# Conclusion



The Purposeful Motion Design Component Library demonstrates how motion can transform user interfaces into clearer, more engaging, and more supportive experiences. By implementing eight reusable components in HTML, CSS, and JavaScript, this project highlights motion's role in providing feedback, guiding attention, and adding delight.

Through timing, easing, and staging, each animation was designed to balance usability with enjoyment: loaders reassure during waiting, checkmarks and confetti celebrate success, shakes highlight errors, and modals or cards guide focus without confusion.

Combined with competitor research and storyboard applications, this library shows how purposeful motion design is both practical and adaptable across real-world interfaces. It serves as a foundation for building design systems that are not only functional but also enjoyable to use.